

Material decay & condition survey

What this tool produces · *A localized, metric diagnostic report mapping specific material degradations (e.g., cracking, surface erosion) directly onto the architectural geometry of the historical asset,.*

G1 · Proof of Impact

Dimensions

Heritage

Environmental

Spatial

Preparing the workshop · *Material decay & condition survey*

G1 · Proof of Impact

Why this tool matters

Listening to the scars of a building; recognizing that every crack, stain, and eroded brick tells the story of time and weather, demanding careful healing rather than violent erasure.

Evidence we produce

A localized, metric diagnostic report mapping specific material degradations (e.g., cracking, surface erosion) directly onto the architectural geometry of the historical asset,.

Duration & Material

Duration · 2 to 4 weeks of field survey and data compilation

Material · Spherical cameras, non-destructive testing equipment, HBIM/GIS software, standardized condition survey grids

Reference case

The Carta del Rischio project in Mirandola (Italy) surveyed an entire urban block heavily damaged by an earthquake. Using spherical photogrammetry integrated into a GIS environment, researchers rapidly mapped the exact condition, metric deformations, and vulnerabilities of building facades. This allowed operators to continuously monitor the post-seismic reconstruction and prioritize preservation interventions remotely,.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Element × condition state · severity matrix

Duration
2 to 4 weeks of field survey and data
compilation

Element × condition state

Element	Sound	Surface decay	Structural decay	Loss	Action
Stone	☐	☐	☐	☐	
Brick / Mortar	☐	☐	☐	☐	
Wood	☐	☐	☐	☐	
Metal	☐	☐	☐	☐	
Render / Plaster	☐	☐	☐	☐	
Roofing	☐	☐	☐	☐	
Glass	☐	☐	☐	☐	
Decoration	☐	☐	☐	☐	

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.